Statistical Validation Theory

Bias, Variance, MSE, Coverage, Type I Error, and Power

A Beginner-Friendly Guide with Full Mathematics and Worked Examples

Applied to the Spatial Omics Measurement Error Framework

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Prerequisites: Core Concepts You Need First

WHAT IS AN ESTIMATOR?

An **estimator** is a recipe (formula) that takes data and produces a guess for some unknown quantity.

In our research: the unknown quantity is the **true CCI score** T_{lr}^{true} (how strongly ligand l and receptor r interact in the tissue). We cannot observe this directly because we only measured some genes. Our estimator $\hat{T}_{lr}$ uses imputed (reconstructed) gene expression data to guess T_{lr}^{true}.

The naive estimator: Compute CCI directly from imputed data, ignoring the fact that imputation introduced errors.

$$\hat{T}_{lr}^{\text{naive}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \hat{\lambda}_{il} \cdot \hat{\lambda}_{jr}$$

The corrected estimator: Subtract the estimated bias (our regression calibration).

$$\hat{T}_{lr}^{\text{corrected}} = \hat{T}_{lr}^{\text{naive}} - \hat{\rho}_{lr}\sqrt{\widehat{\text{MSE}}(l) \cdot \widehat{\text{MSE}}(r)}$$

Analogy — A Bathroom Scale: Imagine your bathroom scale consistently reads 2 kg too high. Your *true weight* is the unknown. The scale reading is the *naive estimator*. Subtracting 2 kg is the *corrected estimator*. The "2 kg" is the *bias*. But the scale also fluctuates randomly by ± 0.3 kg each time — that's the *variance*. Our goal: find an estimator with small bias AND small variance.

WHAT IS A SIMULATION WITH KNOWN GROUND TRUTH?

In real data, we never know the true CCI score — that's why we need estimators. But in a **simulation**, WE create the data, so WE know the truth. This lets us:

1. Generate fake cells with known expression rates $\mathbf{\Lambda}$ (we choose these)
2. Simulate Poisson counts from these rates
3. Pretend we only measured some genes (panel selection)
4. Run our imputation + CCI pipeline
5. Compare our estimates $\hat{T}_{lr}$ to the truth T_{lr}^{true} (which we know because we created the data)
6. Repeat 1000+ times to see how the estimator behaves on average

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Part 1: Bias of the Corrected Estimator Under Simulation

1.1 What Is Bias? (The Simplest Possible Explanation)

DEFINITION OF BIAS

The **bias** of an estimator $\hat{\theta}$ for a true parameter θ is:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

In words: **bias = average estimate minus truth**.

- If Bias > 0: the estimator *overestimates* on average
- If Bias $= 0$: the estimator is *unbiased* (correct on average)
- If Bias < 0: the estimator *underestimates* on average

Why "On Average" Matters

A single estimate can be off in any direction — that's random error. Bias is about the *systematic* direction. If you weigh yourself 1000 times on a biased scale, the *average* of those 1000 readings will be off by exactly the bias. Individual readings will scatter around that wrong average.

An unbiased estimator might give you 48, 53, 47, 51, 52, ... for a true value of 50. The average is ~50 (correct). A biased estimator gives 53, 58, 52, 56, 57, ... The average is ~55 (systematically 5 too high).

1.2 Bias in Our Specific Problem

Recall from the bias derivation: when we compute CCI on imputed data, the expected value of the naive estimator is:

$$\mathbb{E}[\hat{T}_{lr}^{\text{naive}}] = T_{lr}^{\text{true}} + \underbrace{\rho_{lr}\sqrt{\text{MSE}(l) \cdot \text{MSE}(r)}}_{\text{bias}>0}$$

So:

$$\text{Bias}(\hat{T}_{lr}^{\text{naive}}) = \rho_{lr}\sqrt{\text{MSE}(l) \cdot \text{MSE}(r)} > 0$$

The naive estimator is **positively biased** — it makes interactions look stronger than they really are.

1.3 Bias of the Corrected Estimator

Our corrected estimator subtracts an *estimate* of the bias:

$$\hat{T}_{lr}^{\text{corrected}} = \hat{T}_{lr}^{\text{naive}} - \hat{b}_{lr}$$

where $\hat{b}_{lr} = \hat{\rho}_{lr}\sqrt{\widehat{\text{MSE}}(l) \cdot \widehat{\text{MSE}}(r)}$ is our bias estimate.

BIAS OF THE CORRECTED ESTIMATOR

$$\begin{aligned}\text{Bias}(\hat{T}_{lr}^{\text{corrected}}) &= \mathbb{E}[\hat{T}_{lr}^{\text{corrected}}] - T_{lr}^{\text{true}} \\ &= \underbrace{\mathbb{E}[\hat{T}_{lr}^{\text{naive}}] - T_{lr}^{\text{true}}}_{\text{original bias } b_{lr}} - \underbrace{\mathbb{E}[\hat{b}_{lr}]}_{\text{estimated bias}} \\ &= b_{lr} - \mathbb{E}[\hat{b}_{lr}]\end{aligned}$$

In words: the corrected estimator's bias equals the *original bias* minus the *average of our bias estimate*.

- If $\mathbb{E}[\hat{b}_{lr}] = b_{lr}$ (perfect bias estimation): corrected bias = 0. **Ideal case.**
- If $\mathbb{E}[\hat{b}_{lr}] < b_{lr}$ (underestimate bias): corrected bias > 0. **Under-correction — some false positives remain.**
- If $\mathbb{E}[\hat{b}_{lr}] > b_{lr}$ (overestimate bias): corrected bias < 0. **Over-correction — may lose true interactions.**

1.4 How to Measure Bias in Simulation

Step 1: Set up the ground truth.

Choose a true rate matrix $\mathbf{\Lambda}$ (e.g., $N = 5000$ cells, $G = 500$ genes, $K = 10$ programs). Compute the true CCI scores T_{lr}^{true} for all LR pairs using the full, noise-free $\mathbf{\Lambda}$.

Step 2: Simulate one experiment.

Generate counts $y_{ig} \sim \text{Poisson}(\lambda_{ig})$. Select panel S (300 genes). Impute the missing 200 genes. Compute $\hat{T}_{lr}^{\text{naive}}$ and $\hat{T}_{lr}^{\text{corrected}}$.

Step 3: Repeat R times (e.g., $R = 1000$).

Each repetition r gives different Poisson draws, so different counts, different imputations, different CCI estimates. Collect $\{\hat{T}_{lr}^{(1)}, \hat{T}_{lr}^{(2)}, \ldots, \hat{T}_{lr}^{(R)}\}$ for both naive and corrected.

Step 4: Compute empirical bias.

$$\widehat{\text{Bias}}_{\text{naive}} = \frac{1}{R}\sum_{r=1}^{R} \hat{T}_{lr}^{\text{naive},(r)} - T_{lr}^{\text{true}}$$

$$\widehat{\text{Bias}}_{\text{corrected}} = \frac{1}{R}\sum_{r=1}^{R} \hat{T}_{lr}^{\text{corrected},(r)} - T_{lr}^{\text{true}}$$

1.5 Complete Worked Example

Example 1.1 — Bias Measurement for One LR Pair

Setup: True CCI score for CXCL12–CXCR4 interaction: $T^{\text{true}} = 2.50$.

We run $R = 1000$ simulations. Here are the results:

	Naive Estimator	Corrected Estimator
Simulation 1	$\hat{T}^{(1)} = 3.21$	$\hat{T}_c^{(1)} = 2.68$
Simulation 2	$\hat{T}^{(2)} = 2.98$	$\hat{T}_c^{(2)} = 2.41$
Simulation 3	$\hat{T}^{(3)} = 3.45$	$\hat{T}_c^{(3)} = 2.89$
$\vdots$	$\vdots$	$\vdots$
Simulation 1000	$\hat{T}^{(1000)} = 3.12$	$\hat{T}_c^{(1000)} = 2.55$

Compute averages:

$$\bar{\hat{T}}^{\text{naive}} = \frac{1}{1000}\sum_{r=1}^{1000}\hat{T}^{(r)} = 3.14$$

$$\bar{\hat{T}}^{\text{corrected}} = \frac{1}{1000}\sum_{r=1}^{1000}\hat{T}_c^{(r)} = 2.58$$

Compute biases:

$$\text{Bias}_{\text{naive}} = 3.14 - 2.50 = \mathbf{+0.64} \quad \text{(overestimates by 0.64)}$$

$$\text{Bias}_{\text{corrected}} = 2.58 - 2.50 = \mathbf{+0.08} \quad \text{(overestimates by only 0.08)}$$

Bias reduction:

$$\text{Reduction} = 1 - \frac{|\text{Bias}_{\text{corrected}}|}{|\text{Bias}_{\text{naive}}|} = 1 - \frac{0.08}{0.64} = \mathbf{87.5\%}$$

Interpretation: The correction removed 87.5% of the bias. The remaining 0.08 is *residual bias* — our bias estimate $\hat{b}_{lr}$ slightly underestimated the true bias of 0.64 (it estimated ~0.56 on average).

1.6 When Is Residual Bias Acceptable?

RELATIVE BIAS

To judge whether bias matters, compare it to the standard deviation:

$$\text{Relative Bias} = \frac{|\text{Bias}(\hat{T})|}{\text{Std}(\hat{T})}$$

- Relative bias < 0.1: bias is negligible (dominated by random variation)
- Relative bias $0.1 - 0.5$: bias is present but may be acceptable
- Relative bias > 0.5: bias is the dominant source of error — must correct

Example 1.2 — Relative Bias Assessment

Continuing from Example 1.1:

- Naive: $|\text{Bias}| = 0.64$, $\text{Std} = 0.42$, Relative Bias $= 0.64/0.42 = 1.52$ — **bias dominates**
- Corrected: $|\text{Bias}| = 0.08$, $\text{Std} = 0.48$, Relative Bias $= 0.08/0.48 = 0.17$ — **bias is minor**

The naive estimator's error is primarily systematic (bias). The corrected estimator's error is primarily random (variance). This is exactly what we want — random error averages out with more data; systematic error does not.

Part 2: Variance and Mean Squared Error Comparison

2.1 What Is Variance? (From Absolute Basics)

VARIANCE EXPLAINED

The **variance** measures how much an estimator *fluctuates* across repeated experiments.

$$\text{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]$$

In words: variance = average squared distance from the estimator's own average.

Standard deviation $= \sqrt{\text{Var}}$ — has the same units as the estimator, easier to interpret.

Analogy — Archery:

- **Low bias, low variance:** arrows clustered tightly around the bullseye. *Ideal*.
- **Low bias, high variance:** arrows scattered widely but centered on the bullseye. *Unbiased but imprecise*.
- **High bias, low variance:** arrows clustered tightly but away from the bullseye. *Precise but wrong*.
- **High bias, high variance:** arrows scattered widely and off-center. *Worst case*.

2.2 Why Correction Can Increase Variance

Our corrected estimator is:

$$\hat{T}^{\text{corr}} = \hat{T}^{\text{naive}} - \hat{b}$$

The variance of the corrected estimator:

VARIANCE OF THE CORRECTED ESTIMATOR

$$\text{Var}(\hat{T}^{\text{corr}}) = \text{Var}(\hat{T}^{\text{naive}}) + \text{Var}(\hat{b}) - 2\,\text{Cov}(\hat{T}^{\text{naive}}, \hat{b})$$

Key insight: if the bias estimate $\hat{b}$ is noisy (high $\text{Var}(\hat{b})$), the correction *adds* variance. This is the **bias-variance tradeoff** — reducing bias can increase variance.

2.3 Mean Squared Error: The Complete Picture

THE MSE DECOMPOSITION (MOST IMPORTANT FORMULA IN STATISTICS)

$$\boxed{\text{MSE}(\hat{\theta}) = \text{Bias}^2(\hat{\theta}) + \text{Var}(\hat{\theta})}$$

In words: **total error = systematic error + random error**.

MSE answers the question: "On average, how far is my estimate from the truth?" It accounts for BOTH bias and variance simultaneously.

Why MSE, not just bias or variance alone?

- An estimator with zero bias but enormous variance is useless (individual estimates wildly wrong)
- An estimator with zero variance but large bias is also useless (consistently wrong by the same amount)
- MSE captures both — the estimator with lowest MSE is "best" in a squared-error sense

2.4 How to Compute MSE in Simulation

EMPIRICAL MSE COMPUTATION

From R simulation replicates:

$$\widehat{\mathrm{MSE}} = \frac{1}{R}\sum_{r=1}^{R}\left(\hat{T}^{(r)} - T^{\mathrm{true}}\right)^2$$

This can be decomposed as:

$$\widehat{\mathrm{MSE}} = \underbrace{\left(\frac{1}{R}\sum_{r=1}^{R}\hat{T}^{(r)} - T^{\mathrm{true}}\right)^2}_{\widehat{\mathrm{Bias}}^2} + \underbrace{\frac{1}{R}\sum_{r=1}^{R}\left(\hat{T}^{(r)} - \frac{1}{R}\sum_{r'}\hat{T}^{(r')}\right)^2}_{\widehat{\mathrm{Var}}}$$

2.5 Complete Worked Example: Naive vs. Corrected

Example 2.1 — Full MSE Comparison

Setup: True CCI score $T^{\mathrm{true}} = 2.50$. Results from $R = 1000$ simulations:

Metric	Naive Estimator	Corrected Estimator	Winner
Mean of estimates	$\bar{\hat{T}} = 3.14$	$\bar{\hat{T}}_c = 2.58$	
Bias	$3.14 - 2.50 = +0.64$	$2.58 - 2.50 = +0.08$	Corrected
Bias²	$0.64^2 = 0.410$	$0.08^2 = 0.006$	Corrected
Variance	$0.42^2 = 0.176$	$0.48^2 = 0.230$	Naive
MSE = Bias² + Var	$0.410 + 0.176 = \mathbf{0.586}$	$0.006 + 0.230 = \mathbf{0.236}$	**Corrected**
RMSE = √MSE	$\sqrt{0.586} = 0.765$	$\sqrt{0.236} = 0.486$	Corrected

Interpretation:

- The corrected estimator has **higher variance** (0.230 vs 0.176) — the price of correction
- But **much lower bias²** (0.006 vs 0.410) — the benefit of correction
- **Net effect: MSE drops 60%** (0.586 → 0.236). The bias reduction more than compensates for the variance increase.
- The typical error (RMSE) is 0.77 for naive vs 0.49 for corrected — **36% smaller errors**

2.6 The MSE Ratio: When Is Correction Worth It?

MSE RATIO TEST

$$\text{MSE Ratio} = \frac{\text{MSE}(\hat{T}^{\text{corrected}})}{\text{MSE}(\hat{T}^{\text{naive}})}$$

- Ratio < 1: correction improves things — **use corrected estimator**
- Ratio $= 1$: no change — correction neither helps nor hurts
- Ratio > 1: correction makes things worse — **keep naive estimator**

The correction is beneficial when the bias² saved exceeds the variance added:

$$\text{Correction helps when: } \text{Bias}^2_{\text{naive}} - \text{Bias}^2_{\text{corr}} > \text{Var}(\hat{b})$$

2.7 Efficiency

RELATIVE EFFICIENCY

The **relative efficiency** of estimator A compared to B:

$$\text{RE}(A, B) = \frac{\text{MSE}(B)}{\text{MSE}(A)}$$

RE > 1 means A is more efficient (lower MSE). In our example:

$$\text{RE}(\text{corrected, naive}) = \frac{0.586}{0.236} = 2.48$$

The corrected estimator is **2.48 times more efficient** — you would need 2.48× as many samples with the naive estimator to achieve the same MSE as the corrected estimator.

Part 3: Coverage Probability of Confidence Intervals

3.1 What Is a Confidence Interval? (The Correct Interpretation)

CONFIDENCE INTERVAL DEFINITION

A $100(1-\alpha)\%$ **confidence interval** for parameter θ is a random interval $[L, U]$ (computed from data) such that:

$$P(\theta \in [L, U]) = 1 - \alpha$$

For a 95% CI ($\alpha = 0.05$): if you repeated the experiment infinitely many times and computed a CI each time, **95% of those intervals would contain the true value**.

Common Misconception

A 95% CI does NOT mean "there is a 95% probability that the true value is in this interval." The true value is fixed (not random). The interval is random (it changes with each experiment). The probability statement is about the *procedure*, not any single interval.

Analogy — Fishing Net: You throw a net (CI) toward a fish (true value). Before throwing, there's a 95% chance the net lands around the fish. After throwing, the fish is either caught or not — the 95% described the accuracy of your throwing technique, not the state of any particular throw.

3.2 CI for the CCI Statistic

From our panel perturbation stability analysis, we get B estimates $\{\hat{T}^{(1)}, \ldots, \hat{T}^{(B)}\}$. The **percentile confidence interval**:

PERCENTILE CONFIDENCE INTERVAL

$$\mathrm{CI}_{95\%} = \left[\hat{T}^{(\lfloor 0.025 \cdot B \rfloor)},\ \hat{T}^{(\lceil 0.975 \cdot B \rceil)}\right]$$

where $\hat{T}^{(k)}$ is the k-th smallest value (order statistic).

Alternatively, using the **normal approximation**:

$$\mathrm{CI}_{95\%} = \hat{T} \pm 1.96 \cdot \widehat{\mathrm{SE}}$$

where $\widehat{\mathrm{SE}} = \mathrm{Std}(\hat{T}^{(1)}, \ldots, \hat{T}^{(B)})$.

3.3 What Is Coverage Probability?

COVERAGE PROBABILITY

The **coverage probability** is the fraction of simulated CIs that actually contain the true value:

$$\text{Coverage} = \frac{1}{R}\sum_{r=1}^{R} \mathbb{1}\left[T^{\text{true}} \in \text{CI}^{(r)}\right]$$

where $\text{CI}^{(r)}$ is the confidence interval from the r-th simulation.

- For a nominal 95% CI: target coverage = 0.95
- Coverage < 0.95: CIs are too narrow (**anti-conservative**, overconfident)
- Coverage > 0.95: CIs are too wide (**conservative**, underconfident)

3.4 Complete Worked Example

Example 3.1 — Coverage Probability Simulation

Setup: True CCI score $T^{\text{true}} = 2.50$. We run $R = 1000$ simulations. In each, we compute a 95% CI using both the naive and corrected estimators.

Three sample simulations (out of 1000):

Sim #	Naive CI	Contains 2.50?	Corrected CI	Contains 2.50?
1	$[2.38, 4.04]$	Yes	$[1.74, 3.62]$	Yes
2	$[2.55, 3.41]$	**No** (2.50 below)	$[1.98, 2.84]$	Yes
3	$[2.72, 3.98]$	**No** (2.50 below)	$[2.19, 3.45]$	Yes

Notice: the naive CIs are shifted upward (because the estimator is biased), so they often miss the true value by being entirely *above* it.

Results across all 1000 simulations:

Metric	Naive 95% CI	Corrected 95% CI
Actual coverage	$\mathbf{782/1000 = 78.2\%}$	$\mathbf{938/1000 = 93.8\%}$
Target coverage	95%	95%
Under-coverage	$95 - 78.2 = 16.8\%$	$95 - 93.8 = 1.2\%$
Average CI width	1.64	1.88
CIs entirely above truth	198 (19.8%)	41 (4.1%)
CIs entirely below truth	20 (2.0%)	21 (2.1%)

Interpretation:

- The naive CI has **78.2% coverage instead of 95%** — catastrophically anti-conservative. It misses the true value 21.8% of the time, not the expected 5%.
- The missed cases are almost all above the truth (19.8%) — the bias shifts the CI upward.
- The corrected CI achieves **93.8% coverage** — close to the target 95%. The slight under-coverage (1.2%) is due to residual bias.

- The corrected CI is slightly wider (1.88 vs 1.64) — the price of honesty. But it's *honest*: when it says 95%, it means ~94%.

3.5 Formal Test of Coverage

TEST WHETHER COVERAGE EQUALS THE NOMINAL LEVEL

Under the null H_0: true coverage $= 1 - \alpha = 0.95$, the number of "hits" (CIs containing the truth) follows $\text{Binomial}(R, 0.95)$.

The test statistic:

$$Z = \frac{\hat{p} - 0.95}{\sqrt{0.95 \times 0.05/R}}$$

where $\hat{p}$ is the observed coverage proportion. Reject if $|Z| > 1.96$.

For our example (naive): $Z = (0.782 - 0.95)/\sqrt{0.95 \times 0.05/1000} = -0.168/0.00689 = -24.4$. **Massively significant** — coverage is far below nominal.

For corrected: $Z = (0.938 - 0.95)/\sqrt{0.0000475} = -0.012/0.00689 = -1.74$. **Not significant** at $\alpha = 0.05$ — coverage is statistically consistent with 95%.

Part 4: Type I Error Rate and Power

4.1 Hypothesis Testing: The Basics

THE TWO HYPOTHESES

For each LR pair, we test:

- H_0 **(null):** There is no spatial interaction between ligand l and receptor r. The CCI score is no larger than expected by chance.
- H_1 **(alternative):** There IS a spatial interaction. The CCI score is significantly larger than chance.

We compute a p-value and reject H_0 if $p < 0.05$ (declaring the interaction "significant").

4.2 The Four Outcomes

	H_0 is actually TRUE (no real interaction)	H_0 is actually FALSE (real interaction exists)
We reject H_0 (declare significant)	**Type I Error** (False Positive) "We said it's real, but it's not"	**Correct!** (True Positive) "We correctly found it"
We fail to reject H_0 (declare not significant)	**Correct!** (True Negative) "We correctly ignored it"	**Type II Error** (False Negative) "We missed a real one"

KEY RATES

$$\text{Type I Error Rate} = \alpha = P(\text{reject } H_0 \mid H_0 \text{ is true}) \quad (\text{target: } 0.05)$$

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ is true}) \quad (\text{want: as high as possible})$$

$$\text{Type II Error Rate} = \beta = 1 - \text{Power} = P(\text{fail to reject } H_0 \mid H_1 \text{ is true})$$

Analogy — Criminal Trial:

- H_0: defendant is innocent
- H_1: defendant is guilty
- Type I Error: convicting an innocent person (false positive) — the system tries hard to avoid this
- Type II Error: acquitting a guilty person (false negative) — unfortunate but less "bad" than convicting an innocent
- Power: probability of correctly convicting a guilty person

4.3 Why Imputation Bias Inflates Type I Error

The Core Problem in Our Research

When the CCI test statistic is biased upward, the p-values are **too small** under the null. Pairs with no real interaction get small p-values and are falsely declared significant.

If nominal $\alpha = 0.05$ but actual Type I Error is 0.25, then **25% of null interactions are falsely discovered** instead of the expected 5%.

4.4 How to Measure Type I Error in Simulation

TYPE I ERROR RATE ESTIMATION

Key idea: Simulate data where H_0 is TRUE (no spatial interaction), then check how often the test incorrectly rejects.

Procedure:

1. Generate data with *independent* ligand and receptor expression (no spatial co-patterning). Specifically: assign expression values to cells *independently* of their spatial locations.
2. Run the full pipeline (imputation → CCI → permutation test)
3. Record whether $p < 0.05$
4. Repeat R times

$$\widehat{\alpha} = \frac{1}{R} \sum_{r=1}^{R} \mathbb{1}[p^{(r)} < 0.05]$$

Target: $\widehat{\alpha} \approx 0.05$. If $\widehat{\alpha} \gg 0.05$, the test has inflated Type I error.

4.5 How to Measure Power

POWER ESTIMATION

Key idea: Simulate data where H_1 is TRUE (real interaction), then check how often the test correctly rejects.

Procedure:

1. Generate data with a *real* spatial CCI pattern of known strength δ (e.g., ligand-high cells are spatially adjacent to receptor-high cells)
2. Run the full pipeline
3. Record whether $p < 0.05$
4. Repeat R times

$$\widehat{\text{Power}} = \frac{1}{R} \sum_{r=1}^{R} \mathbb{1}[p^{(r)} < 0.05]$$

Compute for different effect sizes δ to get a **power curve**.

4.6 Complete Worked Example

Example 4.1 — Type I Error and Power Comparison

Type I Error (under H_0, 1000 simulations):

Test	Rejections at $\alpha = 0.05$	Empirical Type I Error	Status
Naive (no correction)	237 / 1000	**23.7%**	**INFLATED** (4.7× nominal)
Corrected (our method)	58 / 1000	**5.8%**	**Close to nominal**
Full data (oracle)	47 / 1000	**4.7%**	Benchmark

Formal test of Type I error control:

Under H_0: $\hat{\alpha} \sim \text{Binomial}(1000, 0.05)/1000$. The 95% acceptance region for a valid test is $[0.036, 0.064]$.

- Naive: 0.237 — **far outside**. Test is invalid.
- Corrected: 0.058 — **inside**. Test is valid.
- Oracle: 0.047 — inside. (As expected for the gold standard.)

Power (under H_1 with effect size δ, 1000 simulations per δ):

Effect Size (δ)	Power: Naive	Power: Corrected	Power: Oracle
0.0 (null)	0.237*	0.058	0.047
0.5 (weak)	0.512	0.289	0.320
1.0 (moderate)	0.834	0.671	0.715
2.0 (strong)	0.978	0.943	0.961
3.0 (very strong)	0.999	0.994	0.997

** The naive "power" at $\delta = 0$ is its Type I error rate, not true power.*

Interpretation:

- The naive estimator has **higher apparent power at every effect size** — but this is an *illusion*. Its high "power" at $\delta = 0$ (23.7%) shows it's rejecting too aggressively. It's not more sensitive; it's more *reckless*.
- The corrected estimator has **proper Type I error control** (5.8%) with only modest power loss vs. the oracle (67.1% vs 71.5% at $\delta = 1$).
- At strong effects ($\delta \geq 2$), the corrected and oracle estimators are nearly identical — the correction barely costs any true detections.
- **The correction's price:** ~4% lower power at moderate effects. **The correction's benefit:** 18% fewer false positives. This is an excellent tradeoff.

4.7 The Power-Type I Error Tradeoff (ROC-Like Analysis)

SIZE-ADJUSTED POWER COMPARISON

Since the naive test has inflated Type I error, a fair comparison adjusts for this. Set the rejection threshold so both tests have *exactly* 5% Type I error, then compare power.

For the naive estimator, achieve $\alpha = 0.05$ by using threshold $p < 0.008$ (instead of $p < 0.05$).

Size-adjusted power at $\delta = 1.0$:

- Naive (at $p < 0.008$): 0.52
- Corrected (at $p < 0.05$): 0.67

After size-adjustment, the **corrected estimator has HIGHER power**. The naive estimator's apparent power advantage was entirely due to its inflated false positive rate.

Part 5: Performance as a Function of Sample Size and Noise

5.1 What Are "Sample Size" and "Noise" in Our Context?

PARAMETERS THAT AFFECT PERFORMANCE

Parameter	Symbol	What It Controls	Range in Our Experiments
Number of cells	N	How many cells in the tissue section	500 – 500,000
Panel size	m	How many genes are measured	100 – 5,000
Expression level	$\bar{\lambda}$	Average counts per gene (affects Poisson noise)	0.1 – 200
Latent rank	K	Number of gene programs (biological complexity)	10 – 60
Spatial smoothness	ρ	Moran's I of imputation errors	0 – 0.9

5.2 How Bias Behaves with Parameters

BIAS AS A FUNCTION OF PARAMETERS

$$\text{Bias}_{\text{naive}} = \rho_{lr}\sqrt{\text{MSE}(l) \cdot \text{MSE}(r)}$$

Substituting the MSE bound $\text{MSE}(g) \approx C \cdot K\lambda_{\max}/m$:

$$\text{Bias}_{\text{naive}} \approx \rho_{lr} \cdot C \cdot \frac{K\lambda_{\max}}{m}$$

How each parameter affects bias:

Parameter Change	Effect on Bias	Intuition
$m \uparrow$ (more genes measured)	Bias $\downarrow$ as $1/m$	Better imputation → smaller errors → less bias

$K \uparrow$ (more programs)	Bias $\uparrow$ linearly	More complex biology → harder imputation → larger errors
$\rho \uparrow$ (more spatial smoothing)	Bias $\uparrow$ linearly	More aggressive spatial methods → more correlated errors
$\lambda_{\max} \uparrow$ (higher expression)	Bias $\uparrow$ linearly	Poisson variance grows with rate → larger absolute errors
$N \uparrow$ (more cells)	Bias unchanged (to first order)	Bias is per-edge; more cells adds more edges but same bias per edge

5.3 How Variance Behaves with Parameters

VARIANCE AS A FUNCTION OF PARAMETERS

$$\mathrm{Var}(\hat{T}_{lr}) \approx \frac{1}{|\mathcal{E}|}\left[\bar{\lambda}_l^2 \cdot \mathrm{MSE}(r) + \bar{\lambda}_r^2 \cdot \mathrm{MSE}(l) + \mathrm{MSE}(l) \cdot \mathrm{MSE}(r)\right]$$

Key scaling: $|\mathcal{E}| \propto N$ (more cells → more edges).

Parameter Change	Effect on Variance	Intuition
$N \uparrow$	Variance $\downarrow$ as $1/N$	More edges → averaging over more pairs → less noise
$m \uparrow$	Variance $\downarrow$ as $1/m$	Better imputation → smaller gene-level errors
$\bar{\lambda} \uparrow$	Variance $\uparrow$	Higher expression → larger absolute products → more variable products

5.4 How MSE Behaves: The Complete Picture

MSE SCALING LAWS

$$\text{MSE}_{\text{naive}} = \underbrace{\rho_{lr}^2 \cdot C^2 \cdot \frac{K^2 \lambda_{\max}^2}{m^2}}_{\text{Bias}^2} + \underbrace{\frac{1}{N} \cdot g(m, \bar{\lambda}, K)}_{\text{Var}}$$

$$\text{MSE}_{\text{corrected}} = \underbrace{\text{small residual}}_{\text{Bias}^2 \approx 0} + \underbrace{\frac{1}{N} \cdot g(m, \bar{\lambda}, K) + \frac{\text{Var}(\hat{b})}{1}}_{\text{Var (slightly higher)}}$$

5.5 Complete Worked Example: Performance Table

Example 5.1 — Performance Across Conditions

Fixed parameters: $K = 30$, $\bar{\lambda} = 20$, $\rho = 0.5$, effect size $\delta = 1.0$.

Each cell shows: MSE (Naive) → MSE (Corrected). **Bold** = better.

	Panel Size (m)			
Cells (N)	$m = 100$	$m = 300$	$m = 1000$	$m = 5000$
$N = 500$	2.41 → **1.89**	0.87 → **0.52**	0.31 → **0.28**	0.12 → **0.11**
$N = 5{,}000$	1.95 → **0.48**	0.59 → **0.24**	0.15 → **0.09**	0.04 → **0.03**
$N = 50{,}000$	1.82 → **0.19**	0.43 → **0.05**	0.07 → **0.02**	0.01 → **0.01**
$N = 500{,}000$	1.80 → **0.08**	0.41 → **0.01**	0.05 → **0.005**	0.004 → **0.003**

Key patterns:

1. **Increasing N helps the corrected estimator much more than the naive.** At $m = 300$: naive MSE drops from 0.87 to 0.41 (2.1×), but corrected drops from 0.52 to 0.01 (52×). Why? The naive MSE is dominated by bias (which doesn't shrink with N), while the corrected MSE is dominated by variance (which shrinks as $1/N$).
2. **Increasing m helps both, but the naive still has a floor.** Even at $m = 5000$, the naive MSE is 0.004 at $N = 500{,}000$ — still 4× the corrected MSE. The bias floor never reaches zero.
3. **The correction helps most in the large-N, small-m regime** — exactly where spatial omics operates (many cells, limited gene panels). At $N = 50{,}000$ and $m = 300$: MSE ratio = $0.43/0.05 = 8.6\times$ improvement.

5.6 Asymptotic Behavior

WHAT HAPPENS AS $N \to \infty$?

Naive estimator:

$$\text{MSE}_{\text{naive}} \xrightarrow{N\to\infty} \text{Bias}^2 = \rho_{lr}^2 \cdot \frac{C^2 K^2 \lambda_{\max}^2}{m^2}$$

The variance vanishes, but the **bias remains**. The naive estimator converges to the **wrong value**. It is *inconsistent* for the true CCI score (unless $\rho = 0$).

Corrected estimator:

$$\text{MSE}_{\text{corrected}} \xrightarrow{N\to\infty} \text{residual bias}^2 \approx 0$$

Both bias and variance vanish. The corrected estimator converges to the **true value**. It is *consistent*.

This Is the Most Important Result

No matter how much data you collect ($N \to \infty$), the naive CCI estimator **never converges to the truth** when applied to imputed spatial data. It converges to $T^{\text{true}} + \text{bias}$. Only the corrected estimator converges to T^{true}.

This is why the correction is not optional — it's **necessary for valid science**. Without it, more data just gives you more confidence in the wrong answer.

5.7 How Noise Level ($\bar{\lambda}$) Affects Performance

Example 5.2 — Performance Across Expression Levels

Fixed: $N = 10{,}000$, $m = 300$, $K = 30$.

Mean Expression ($\bar{\lambda}$)	Poisson CV	Type I Error (Naive)	Type I Error (Corrected)	Power at $\delta = 1$ (Corrected)
0.5 (very low)	141%	0.089	0.054	0.18
2 (low)	71%	0.156	0.057	0.41
10 (moderate)	32%	0.234	0.052	0.67
50 (high)	14%	0.287	0.048	0.89
200 (very high)	7%	0.312	0.051	0.97

Key patterns:

- **Higher expression → worse naive Type I error.** Counter-intuitive! Higher expression means bigger imputation errors in absolute terms → bigger bias → more false positives.
- **The corrected estimator maintains ~5% Type I error across all expression levels.** The correction adapts because it uses the estimated MSE, which scales with expression.
- **Power increases with expression.** Higher expression → better SNR → easier to detect real interactions.

5.8 Summary: When Does the Correction Help Most?

CONDITIONS WHERE CORRECTION IS MOST VALUABLE

Condition	Why
Small panel ($m \leq 500$)	Large MSE → large bias → large correction needed
Many cells ($N > 10{,}000$)	Variance is already small, bias dominates
Graph-based imputation	High spatial error correlation (ρ) → large bias
High-expression ligands/receptors	Larger absolute MSE → larger bias
Both genes imputed ($l, r \notin S$)	Quadratic bias term

THE BOTTOM LINE

For the typical spatial omics experiment ($N = 50{,}000$ cells, $m = 300$ gene panel, graph-based imputation):

- Without correction: **~24% false positive rate** (4.7× the nominal 5%)
- With correction: **~5.5% false positive rate** (close to nominal)
- Power loss from correction: **~4%** at moderate effect sizes
- MSE improvement: **8–10× lower**
- Coverage: **~94% vs ~78%** for 95% nominal CIs

The correction is not just a theoretical improvement — it is **necessary for valid biological conclusions** from imputed spatial omics data.

Statistical Validation Theory — Measurement Error Framework for Spatial Omics Inference
5 Parts • Beginner-friendly with full derivations, worked examples, and simulation protocols